X and R Charts for a Manufacturing Process

Hard-bake process in semiconductor manufacturing

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1. Problem Statement

A hard-bake process is used in conjunction with photolithography in semiconductor manufacturing. During this process, a resist material is treated to achieve the required properties for manufacturing semiconductor wafers. The flow width of the resist is an important quality characteristic that must be monitored to maintain consistent product quality.

The quality control department of a semiconductor manufacturing facility wants to determine whether the process is operating under statistical control. For this purpose, two statistical process control charts, the **$\bar{X}$ (X-bar) chart** and the **R (Range) chart**, will be constructed.

Twenty-five samples, also called subgroups, were collected when the process was believed to be in control. Each sample contains measurements from five wafers, so the subgroup size is $n = 5$. Samples were collected at one-hour intervals, and the flow width of the resist was measured in microns.

The collected data will be used to calculate the mean and range of each subgroup, estimate the overall process mean and variability, establish the upper and lower control limits, and determine whether the process is statistically stable.

2. Given Information

- Number of subgroups: $m = 25$
- Number of wafers per subgroup: $n = 5$
- Total number of measurements: $25 \times 5 = 125$
- Sampling interval: One hour
- Quality characteristic: Resist flow width
- Measurement unit: Microns
- $\bar{X}$-chart constant: $A_2 = 0.577$
- R-chart constant: $D_3 = 0$
- R-chart constant: $D_4 = 2.114$

The constants are selected from the standard control chart constants table for a subgroup size of five.

3. Objectives

The objectives of this problem are:

1. Calculate the mean and range for each of the 25 subgroups.
2. Calculate the grand mean, $\bar{\bar{X}}$, and average range, $\bar{R}$.
3. Construct the R chart to examine process variability.
4. Calculate the control limits for the R chart.
5. Calculate the control limits for the $\bar{X}$ chart.
6. Construct both control charts and identify any points outside the control limits.
7. Determine whether the process appears to be in statistical control and whether the trial control limits can be used for future monitoring.

4. Required Calculations

For each subgroup i, calculate the sample mean:

$$\bar{X}_i = \frac{1}{n} \sum_{j=1}^{n} X_{ij}$$

Calculate the sample range:

$$R_i = X_{\max,i} - X_{\min,i}$$

Calculate the grand mean:

$$\bar{\bar{X}} = \frac{1}{m} \sum_{i=1}^{m} \bar{X}_i$$

Calculate the average range:

$$\bar{R} = \frac{1}{m}\sum_{i=1}^{m} R_i$$

R-chart control limits

$$UCL_R = D_4\bar{R}$$
$$CL_R = \bar{R}$$
$$LCL_R = D_3\bar{R}$$

X-chart control limits

$$UCL_{\bar{X}} = \bar{\bar{X}} + A_2\bar{R}$$
$$CL_{\bar{X}} = \bar{\bar{X}}$$
$$LCL_{\bar{X}} = \bar{\bar{X}} - A_2\bar{R}$$

5. Interpretation Plan

The R chart will be examined first because the X-chart limits depend on the estimate of process variability. After checking the R chart, the X chart will be examined for changes in the process mean.

Any point outside the control limits indicates a possible special cause of variation and should be investigated. Patterns such as long runs on one side of the centre line or sustained trends should also be checked.

If both charts indicate statistical control and no important unusual patterns are found, the trial control limits may be adopted for monitoring future production, subject to confirming that the process is stable and the sampling assumptions are appropriate.

6. Data Summary

Each subgroup is summarised by its mean (X) and its range (R = max − min).

Sample	Wafer 1	Wafer 2	Wafer 3	Wafer 4	Wafer 5	Mean	Range
1	1.3235	1.4128	1.6744	1.4573	1.6914	1.5119	0.3679
2	1.4314	1.3592	1.6075	1.4666	1.6109	1.4951	0.2517
3	1.4284	1.4871	1.4932	1.4324	1.5674	1.4817	0.1390
4	1.5028	1.6352	1.3841	1.2831	1.5507	1.4712	0.3521
5	1.5604	1.2735	1.5265	1.4363	1.6441	1.4882	0.3706
6	1.5955	1.5451	1.3574	1.3281	1.4198	1.4492	0.2674
7	1.6274	1.5064	1.8366	1.4177	1.5144	1.5805	0.4189
8	1.4190	1.4303	1.6637	1.6067	1.5519	1.5343	0.2447
9	1.3884	1.7277	1.5355	1.5176	1.3688	1.5076	0.3589
10	1.4039	1.6697	1.5089	1.4627	1.5220	1.5134	0.2658
11	1.4158	1.7667	1.4278	1.5928	1.4181	1.5242	0.3509
12	1.5821	1.3355	1.5777	1.3908	1.7559	1.5284	0.4204
13	1.2856	1.4106	1.4447	1.6398	1.1928	1.3947	0.4470
14	1.4951	1.4036	1.5893	1.6458	1.4969	1.5261	0.2422

Sample	Wafer 1	Wafer 2	Wafer 3	Wafer 4	Wafer 5	Mean	Range
15	1.3589	1.2863	1.5996	1.2497	1.5471	1.4083	0.3499
16	1.5747	1.5301	1.5171	1.1839	1.8662	1.5344	0.6823
17	1.3680	1.7269	1.3957	1.5014	1.4449	1.4874	0.3589
18	1.4163	1.3864	1.3057	1.6210	1.5573	1.4573	0.3153
19	1.5796	1.4185	1.6541	1.5116	1.7247	1.5777	0.3062
20	1.7106	1.4412	1.2361	1.3820	1.7601	1.5060	0.5240
21	1.4371	1.5051	1.3485	1.5670	1.4880	1.4691	0.2185
22	1.4738	1.5936	1.6583	1.4973	1.4720	1.5390	0.1863
23	1.5917	1.4333	1.5551	1.5295	1.6866	1.5592	0.2533
24	1.6399	1.5243	1.5705	1.5563	1.5530	1.5688	0.1156
25	1.5797	1.3663	1.6240	1.3732	1.6887	1.5264	0.3224

7. Solution, Step by Step

Step 1 — Start with the R chart

Control limits of the X chart depend on process variability. Unless variability is in control, those limits have little meaning. So the **R chart is built first**.

Step 2 — Averages

Quantity	Value
Grand mean (X)	1.50561
Average range (R)	0.32521

Step 3 — Constants for n = 5 (Appendix Table VI)

A2	D3	D4
0.577	0	2.114

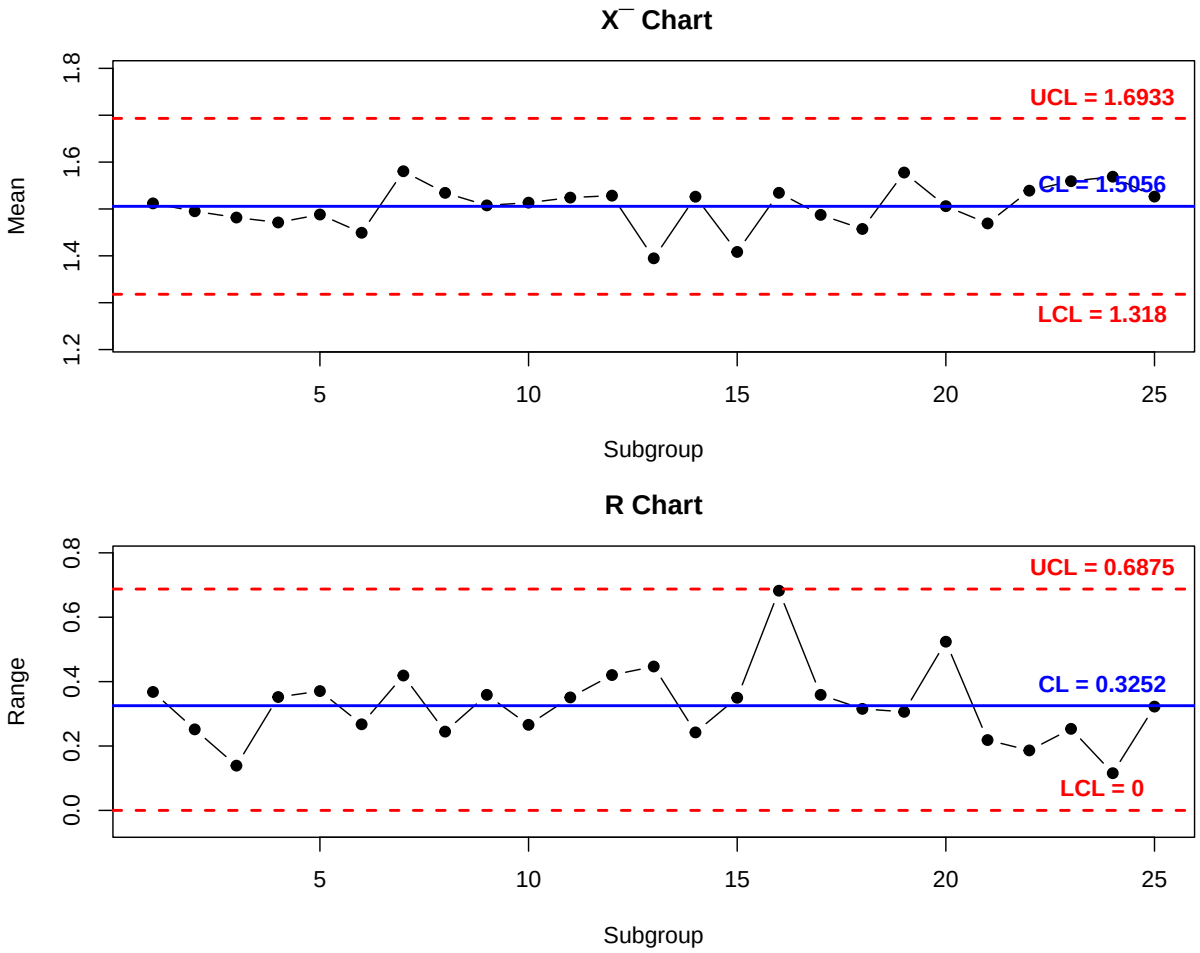
Step 4 — Control limits

R chart: LCL = D · R , CL = R , UCL = D · R

X chart: LCL = X − A · R , CL = X , UCL = X + A · R

Chart	LCL	CL	UCL
R chart	0.000	0.3252	0.6875
X chart	1.318	1.5056	1.6933

Step 5 — Control charts



Step 6 — Out-of-control check

- **R chart:** no points outside the control limits.
- **X chart:** no points outside the control limits.

8. Conclusion

- The **R chart** shows that process variability is in statistical control (the largest range, subgroup 16, stays just below the UCL).
- The **X chart** shows that the process mean is also in statistical control.
- Both charts show **no out-of-control points and no unusual pattern**, so the trial control limits above can be adopted for **monitoring future production** of the hard-bake process.